

P100-058: Differential scanning calorimetry, gel permeation chromatography and linear rheology measurements of poly(D,L-lactide) (Aldrich 805734)

Principia-100 | Standardized portfolio | 1 October 2026

Scenario and evaluation contract

Domain: Polymer physics. The current default task predicts Storage modulus G prime in Pa. Independent unit: whole temperature sweep. Its packaged cohort contains 35 assigned rows in 2 groups.

All supplied outcomes are now exposed. Historical reserved evaluations remain part of the evidence record; future scores are retrospective. No newly established fundamental law or measured deployment impact is admitted by this packaging.

Current portfolio findings

Validated extension: Unfitted PLA3D850 loss prediction is a useful orthogonal response check under frozen storage calibration.

Limit: Loss comes from the same instrument/experiments and does not establish independent replication or fully identified modes.

Unsupported or falsified: Recycled-PLA loss response falsifies general storage-to-loss reliability of the calibrated family.

Limit: Material-specific dissipative behavior or metrology is unresolved; parameters must not be retuned against these exposed outcomes and relabeled fresh.

Versioned tasks

Task family	Target and information	Source scope
original	Storage modulus G prime (Pa)	original corpus
continuation	Storage modulus G prime (Pa)	original corpus
PLA PLA3D850-loss	Unfitted loss modulus (Pa)	supplemental
PLA PLA3D850-storage	Storage modulus (Pa)	supplemental
PLA PLA_recycled-loss	Unfitted loss modulus (Pa)	supplemental
PLA PLA_recycled-storage	Storage modulus (Pa)	supplemental

Different targets, calibration budgets and cohorts have different task IDs. Their raw error values cannot be pooled into a scenario ranking. Exact equations, coefficients, permitted variables, timing, groups and per-model results are bound in the task packages.

Evaluation: the shared CLI provides numerical scoring, explicit coverage and abstention, uncertainty and event diagnostics. Agent review separately assesses scientific support; no aggregate discovery score is produced.

Detailed reference note: Polymer rheology: thermal transfer and its loss-response limit

Research edition: 1 October 2026. Historical results remain preserved; this reader edition uses the shared benchmark evaluator.

Supported findings and constitutive equation

The response is native storage modulus G' in Pa; independently recorded loss modulus G'' is never fitted or used for selection. Whole-temperature development uses 7 sweeps of one PDLA15k batch. The existing T70/T80 cohort is now exposed; all revised comparisons there are retrospective.

$$G'(\omega, T) = \sum_j G_j \frac{(\omega a_T \tau_j)^2}{1 + (\omega a_T \tau_j)^2}.$$

$$G''(\omega, T) = \sum_j G_j \frac{\omega a_T \tau_j}{1 + (\omega a_T \tau_j)^2},$$

$$\log_{10} a_T = -\frac{C_1(T - T_r)}{C_2 + T - T_r}.$$

The selected PDLA state has $T_r=60$ C, $C_1=12$, $C_2=100$ K and 13 fixed clocks from 10^{-6} to 10^6 seconds. Nonnegative amplitudes and all fold states are in rules.json. Ridge was allowed in nested whole-temperature tuning; the full-development selection uses ridge 0. This is a stronger implementation of established Maxwell/WLF physics, not a new constitutive law.

Positive-response development logMAE is 0.571979; the exposed cohort gives 0.311133 and physical MAE 54.2685 Pa, versus the old 0.404461 and 116.619 Pa. Logs score 32 positive readings; physical errors preserve all 35, including 3 negative native readings. The unfitted loss response gives 0.152957 logMAE, using identical coefficients, versus old 0.256751. Sparse, Arrhenius and fractional competitors do not improve the development primary error.

Fresh material challenge and its failure boundary

Before opening new numeric responses, a separate protocol reserved higher temperatures of two commercial PLA materials from Zenodo 17288444. Lower-temperature storage sweeps supplied declared material calibration: 110-140 C for PLA3D850 and 130-150 C for recycled PLA. Independent reserved sweeps were 150-180 C and 160-170 C respectively. Publisher MD5/local SHA256, filename-only selection, parser header repair, model freeze and first response opening are traceable. These are distinct material measurements from the same authors/instrument, not independent laboratories. This is calibrated physical-family transfer, not zero-shot coefficient transfer.

WLF storage logMAE is 0.709324 versus Arrhenius 1.199815 for PLA3D850 (66 rows/4 temperatures), and 0.379836 versus 0.808621 for recycled PLA (30 rows/2 temperatures). Yet all held-temperature biases are negative, and error grows at hotter temperatures. Unfitted loss logMAE is 0.464169 for PLA3D850 but 1.860375 for recycled PLA: the latter falsifies a generally reliable storage-to-loss transfer claim. Old PDLA zero-shot parameters fail badly on both. No final-test performance was used to revise or promote models.

Value, identifiability and next evidence

The solid contribution is a frozen cross-material challenge plus preserved orthogonal-response failure. Mode conditioning reaches 2.4×10^{16} ; individual clocks, exact band masses and zero-frequency viscosity are not identified. An elastic plateau and very slow tail are indistinguishable in these observations. Compliance remains uncorrected. New creep/relaxation or independently calibrated complex-modulus measurements are needed to distinguish spectra and establish processing relevance. Fresh-challenge storage/loss tables and frozen calibrated models are separate from the old-task candidate.

Reproduction and evaluation

Run `python run.py` to verify frozen predictions. Use `python evaluator/evaluate.py example --output /tmp/submission` and `python evaluator/evaluate.py score --submission /tmp/submission --output /tmp/report --trust-code` in a new output location. This trusted-code replay is not a security sandbox. Evaluator scores exposed targets and does not grant novelty from error. Exact source/sample anchors are in `data/observations.csv.gz`; complete numerical evidence and unfavorable models remain in `evidence/`.

Sources and prior art

- Native PDLA record
- Commercial PLA challenge record
- Generalized versus fractional Maxwell prior work
- Spectrum identification prior work

Targeted primary-source checking is not exhaustive novelty adjudication. Established model-family success is distinguished from a new physical law.